NextGen
OLYMPIAD FOUNDATION

GRADE 5

STEM OLYMPIAD

National Level Examination

SUBJECT CODE			

Total Questions: 50 **Total Marks: 50** **Time: 75 Minutes**

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED TO DO SO

- **Carefully Check Your Name, Class, Roll Number, And Other Required Details On The OMR Answer Sheet.**
- **Do Not Forget To Sign The OMR Sheet. Also Write Your Roll No. On The Question Paper Booklet.**
- **Use Only Black/Blue Ball Pen And HB Pencil To Fill The OMR Answer Sheet.**
- **All Questions Are Compulsory And Carry One Mark Each.**
- **Section A: Mathematics & Quantitative Reasoning (15 Questions); Section B: Science & Scientific Reasoning (15 Questions); Section C: Engineering, Technology & Data (10 Questions); Section D: STEM Olympiad Challenge (10 Questions).**
- **Only OMR Answer Sheets Will Be Evaluated.**
- **No Negative Marking For Incorrect Answers.**
- **You May Use The Space Provided On The Last Page For Rough Work (If Needed).**
- **Digital Devices Like Calculators, Smart Watches, Or Mobile Phones Are Not Allowed.**
- **After The Exam, Submit Both The OMR Sheet And The Question Paper To The Invigilator.**

Sample paper

SECTION A – MATHEMATICS & QUANTITATIVE REASONING

Q1. A school had 48,706 notebooks. It distributed 19,858 notebooks to students and later received 6,975 more notebooks. How many notebooks does the school have now?
A. 34,823
B. 35,823
C. 36,823
D. 37,823

Q2. What is 324 x 25?

A. 7,100
B. 8,000
C. 8,100
D. 8,200

Q3. What is 5,040 divided by 24?

A. 200
B. 210
C. 220
D. 240

Q4. Which fraction is greatest?

A. 3/5
B. 5/8
C. 2/3
D. 7/12

Q5. A ribbon is 8.5m long. Meera uses 2.75m and then another 1.8m. How much ribbon is left?

A. 3.85m
B. 3.95m
C. 4.05m
D. 4.15m

Q6. A rectangle has area 96 cm² and width 8 cm. What is its length?
A. 10 cm
B. 11 cm
C. 12 cm
D. 14 cm

Q7. A cube has edge 4 cm. What is its volume?

A. 16 cm³
B. 32 cm³
C. 48 cm³
D. 64 cm³

Q8. What is the smallest common multiple of 6 and 8?

A. 12
B. 18
C. 24
D. 48

Q9. What is the greatest common factor of 18 and 30?
A. 3
B. 6
C. 9
D. 12

Q10. A train leaves at 09:45 and travels 2 h 35 min. When does it arrive?

A. 12:10
B. 12:20
C. 12:30
D. 12:40

Q11. Which number does not belong to the pattern?
8, 16, 24, 35,40

A. 16
B. 24
C. 35
D. 40

Q12. A bag has 5 red, 3 blue and 2 green balls. What fraction of the balls are blue?

A. 1/5
B. 3/10
C. 1/3
D. 3/5

Q13. The perimeter of a square is 44 cm. What is its side?

A. 9 cm
B. 10 cm
C. 11 cm
D. 12 cm

Q14. Which angle is 135°?

A. Acute
B. Right
C. Obtuse
D. Straight

Q15. A bar graph shows 18, 25, 21 and 16 books read in four months. What is the average per month?

A. 18
B. 19
C. 20
D. 21

Sample paper

SECTION B – SCIENCE & SCIENTIFIC REASONING

Q16. Which sequence correctly shows the path of food through the digestive system?

A. Mouth→stomach→food pipe→small intestine→Large intestine
B. Mouth→foodpipe→stomach→small intestine→large intestine
C. Mouth→small intestine→stomach→large intestine
D. Mouth→stomach→large intestine→small intestine

Q17. Which condition is most important for a seed to germinate?
A. Water, air and suitable temperature
B. Only sunlight
C. Only soil
D. Only fertilizer

Q18. Which process in green leaves uses sunlight to make food?

A. Respiration
B. Photosynthesis
C. Transpiration
D. Germination

Q19. Which substance is most likely to dissolve completely in water?

A. Sand
B. Salt
C. Chalk powder
D. Sawdust

Q20. Which method is best to separate tea leaves from tea?
A. Filtration
B. Magnetism
C. Evaporation
D. Freezing

Q21. What happens to the shadow of a pole as the Sun changes position?

A. Its length and direction can change
B. It stays exactly the same
C. It becomes a sound
D. It turns into light

Q22. Which fuel is a fossil fuel?

A. Coal
B. Sunlight
C. Wind
D. Flowing water

Q23. Which action helps recharge groundwater?

A. Covering all soil with concrete
B. Allowing rainwater to soak into the ground
C. Pumping water continuously
D. Blocking ponds

Q24. Why is the small intestine important?

A. Most digestion and absorption occur there
B. It pumps blood
C. It filters air
D. It makes bones

Q25. Which adaptation helps a cactus conserve water?

A. Broad soft leaves
B. Leaves reduced to spines
C. Webbed feet
D. Gills

Q26. Which natural event is measured using a seismograph?

A. Earthquake
B. Rainfall
C. Wind direction
D. Temperature

Q27. Which simple machine is a seesaw?

A. Lever
B. Pulley
C. Screw
D. Wedge

Q28. A magnet attracts which material strongly?

A. Iron
B. Wood
C. Plastic
D. Glass

Q29. Why does a metal spoon feel hotter than a wooden spoon in hot soup?

A. Metal conducts heat better
B. Wood creates cold
C. Metal is lighter
D. Wood absorbs light

Q30. Which is a renewable source of energy?

A. Natural gas
B. Petroleum
C. Solar energy
D. Coal

SECTION C – ENGINEERING, TECHNOLOGY & DATA

Sample paper

Q31. A model bridge must carry more load. Which change is most useful?

A. Add triangular bracing
B. Remove supports
C. Use longer unsupported spans
D. Wet the material

Q32. A simple circuit has a cell, wires and bulb. The bulb glows only when:

A. the circuit is closed
B. one wire is missing
C. both cell terminals are disconnected
D. the switch is open

Q36. A student tests which soil holds the most water. What should be the same for all soil samples?

A. Amount of soil and water added
B. Type of soil
C. Result recorded
D. Colour of label only

Q37. Which structure is strongest for a tall tower model?

A. Broad base with cross-bracing
B. Very narrow base
C. Heavy top with no supports
D. One thin vertical stick

Q38. A rain gauge is used to measure:

Q33. A solar panel produces less electricity in shade because it receives less:

A. light energy
B. mass
C. friction
D. sound

Q34. A 2 kg load is lifted using a pulley. The main advantage of a fixed pulley is that it:

A. changes the direction of force
B. removes gravity
C. makes the load massless
D. creates energy

Q35. Which measurement best tells how fast a toy car moves?

A. Distance travelled per unit time
B. Its colour
C. Its mass only
D. Number of wheels only

A. rainfall
B. temperature
C. wind speed
D. air pressure

Q39. If a 500 mL bottle fills a tank 12 times, how much water enters the tank?

A. 4 L
B. 5 L
C. 6 L
D. 12 L

Q40. A robot starts facing North, moves 4, turns right and moves 6, turns right and moves 4. Where is it relative to the start?

A. 6 steps East
B. 6 steps West
C. 4 steps North
D. 4 steps South

SECTION D – STEM OLYMPIAD CHALLENGE

Q41. A plant is watered equally but one set receives 8 h light and another 2 h. Which variable is being changed?

A. Amount of light
B. Amount of water
C. Plant type
D. Soil mass

Q42. A water filter removes visible dirt but not all germs. What should be done before drinking if safety is uncertain?

A. Use an appropriate purification method such as boiling
B. Add more dirt
C. Leave it open
D. Mix it with soil

Q43. A 1.2 m board is cut into 4 equal pieces. Each piece is:

A. 20 cm
B. 30 cm
C. 40 cm
D. 60 cm

Q44. A device uses 3 cells of 1.5 V each in series. What is the total voltage?

A. 1.5 V
B. 3.0 V
C. 4.5 V
D. 6.0 V

Q45. Which design reduces friction in a moving cart?
A. Use wheels with smooth axles
B. Drag it on sandpaper
C. Add rough pads to the axles
D. Remove the wheels

Q46. A class records temperatures 24°C, 26°C, 28°C, 30°C at equal times. If the pattern continues, the next reading is:
A. 31°C
B. 32°C
C. 33°C
D. 34°C

Q47. A 20-tooth gear turns a 40-tooth gear. Compared with the first gear, the second gear turns:
A. more slowly
B. at the same speed
C. twice as fast
D. without moving

Q48. Which evidence best shows that air occupies space?

A. An inverted glass traps air and stops water entering fully
B. Air is invisible
C. Wind can feel cool
D. Clouds are white

Q49. A model house needs to stay cooler in summer. Which roof choice helps most?

A. Light-coloured reflective roof with insulation
B. Black metal sheet without insulation
C. Transparent roof
D. Dark roof under direct Sun

Q50. A seedling experiment gives heights 6, 9, 13, 18 cm on four days. The increases are +3, +4, +5. If the pattern continues, the next height is:
A. 22 cm
B. 23 cm
C. 24 cm
D. 25 cm

Sample Paper

Sample paper